

MFN Clauses in Sequential Venture Financing

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Abstract

Accelerator contracts are not negotiated, and they increasingly carry most-favored-nation (MFN) clauses under which the accelerator’s tranche converts on the best terms obtained by any investor entering before the first priced round. This article asks what such a clause does to the round that follows it. In a sequential financing game—accelerator, pre-seed investor, seed investor, entrepreneur—the pre-seed investor’s equilibrium share is independent of the amount it advances: valuations scale with the financing need, ownership stakes do not. The clause then operates entirely through the price that investor sets. Converting at the lower of the two bases, it decides whether it absorbs the tranche or passes it to the priced round, and the two branches obey the same threshold equation at two effective investment costs, I and $I + m$, whose roots satisfy $\pi_1 < \pi_2$. Liquidity needs are thereby partitioned into three regimes: neutrality, bunching at the round price, and taxation of the investment cost. The clause is harmless where it is merely redistributive and costly where it binds, and its cost falls on the firms with the largest interim needs. A pre-seed investor that itself converts at the best of the two prices is weakly dominated, strictly so on large needs: a protection granted to an investor can reduce that investor’s own payoff.

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1 Introduction

Accelerator contracts are not bargained. Y Combinator and Techstars run cohort-based programs of roughly three months that combine early financing with training, mentorship and investor networking, and the terms they offer are uniform across the cohort and imposed on follow-on investors (Zeisberger et al., 2017; Gonzalez-Uribe and Leatherbee, 2018). Until 2022, Y Combinator’s standard deal was \$125,000 for 7% of post-money equity. In January 2022 the accelerator added \$375,000 invested on an *uncapped SAFE with a most-favored-nation provision*: a convertible instrument that automatically adopts the most favourable terms obtained by any other investor between the date of issuance and the next priced equity round.

The clause addresses a problem of sequence. The accelerator commits its capital at admission, before the cohort has produced any information, on terms it publishes in advance and applies uniformly to every company. Investors arriving later choose after observing what each company has become, and they negotiate individually. Nothing prevents such an investor from obtaining a more advantageous basis of calculation than the accelerator’s own—a lower valuation, hence a larger share per dollar advanced—and the accelerator, having already invested and having renounced negotiation, has no instrument with which to respond. This is the commitment asymmetry that McAfee and Schwartz (1994) identify in sequential contracting, and the most-favored-nation clause is the standard remedy: it aligns the accelerator’s conversion terms on whatever the intermediate investor obtains, so that moving first no longer means being treated worst.

Announcing the reform, Y Combinator’s then-president argued that the larger check would remove the immediate pressure to fundraise on unfavourable terms, encourage more founders to apply, and make it more likely that companies reach a successful seed round.

These are equilibrium claims about a multi-stage financing game, not accounting identities. Between the accelerator and the priced round, a company that runs short of cash must raise again, and the investor who covers that interim need—we call it the pre-seed investor—sets a price. The MFN clause makes the accelerator’s effective stake a function of that price: the tranche converts at the lower of the pre-seed valuation and the round valuation. The clause therefore does not merely transfer equity; it changes what the pre-seed investor is solving for. Whether the extra capital improves financing outcomes depends on how that investor responds, and the response is not obvious in either direction.

The question belongs to four literatures that have not met on this margin.

The first is the security-design literature on staged venture financing, which asks which instrument implements the efficient continuation decision when capital arrives in tranches. Gompers (1995) documents staging as the industry’s central control device, and Neher (1999) shows that splitting the investment is what makes the en-

trepreneur’s promise credible in the first place. Convertible securities are the recurring answer: Cornelli and Yosha (2003) obtain them from the entrepreneur’s incentive to window-dress interim performance ahead of an abandonment option, Schmidt (2003) from double-sided moral hazard, Repullo and Suarez (2004) from a security-design problem over multiple stages, and Casamatta (2003) from the joint provision of capital and advice; the allocation of control that accompanies them is characterized by Hellmann (1998), and the learning-and-stopping structure by Bergemann and Hege (2005). In all of these the instrument is the unknown and the number of investors is two. Here the instrument is given by practice—a post-money SAFE, whose conversion basis is fixed at issuance—and the question is what a clause grafted onto it does once a third party prices an intermediate round.

The second is the literature on robust contracting and the dilution of early claims. Admati and Pfleiderer (1994) show that a fixed-fraction contract—the inside investor holds a constant share across rounds—is the unique contract that makes the insider’s payoff immune to the manipulation of later-round pricing; the neutrality result below is its SAFE-era counterpart, in an instrument space where the early claim converts at a *price* rather than carrying a fixed share. What the contracts actually contain is documented by Kaplan and Strömberg (2003, 2004) and, for the link between investor characteristics and terms, by Bengtsson and Sensoy (2011), in the tradition of Sahlman (1990). That documentation predates the instrument studied here: Coyle and Green (2014) trace the displacement of priced seed rounds by convertible instruments, the contractual shift that made a standardized, non-negotiated pre-priced claim the normal entry point into a company’s capital structure and thereby created the sequencing problem this paper studies.

The third is the industrial-organization literature on most-favored-customer clauses. McAfee and Schwartz (1994) show that nondiscrimination clauses protect the early signatories of a principal who contracts sequentially and cannot resist opportunism towards them, which is precisely the commitment asymmetry here: the accelerator invests before any uncertainty resolves, the pre-seed investor after observing the shock. That literature generally finds such clauses price-rigidifying and anticompetitive (Cooper, 1986; Besanko and Lyon, 1993), because the promise to match any better price removes the incentive to cut it. The sign reverses in this environment for a reason specific to the financing chain: within one regime the clause leaves the financing threshold of the chain untouched, and the party it constrains is not a seller facing rivals but an intermediary choosing where to place a claim it does not hold.

The fourth is the literature on the tiering of early-stage investors. Hellmann and Thiele (2015) study the equilibrium interaction between angel and venture capital markets, where a downstream tier holds up an upstream one—structurally the closest financing game to the sequence studied here, with the difference that their instruments are given whereas the instrument is the object of study in this paper. That tiers differ in what they bring, and that the difference is priced, is established by Hellmann and Puri (2002) and Hochberg et al. (2007); accelerators specifically are evaluated by Gonzalez-Urbe and Leatherbee (2018) and Hallen et al. (2020), who find effects

on venture development but treat the program’s terms as fixed. Two features of the environment make those terms worth modelling now: the collapse in the cost of starting a company has compressed the interval between the accelerator and the priced round (Ewens et al., 2018), and the risk that a company fails to raise its next round is a first-order determinant of early-stage financing (Nanda and Rhodes-Kropf, 2017; Lerner and Nanda, 2020). It is exactly that risk—the probability that the seed round does not happen—that the clause displaces between the parties.

Our model has four players and one state variable. An accelerator invests (σ_a, ρ_a) before anything is known; a liquidity shock l then hits; a pre-seed investor may cover the residual need $\sigma_p = l - \sigma_a$ at a valuation of its choosing; the project’s success probability π is revealed; and a seed investor funds the sunk cost I against whatever equity remains. Three results follow. First, the pre-seed investor’s equilibrium share is independent of the amount it advances: the valuation it sets scales one-for-one with the financing need, so across companies with similar characteristics but different liquidity shocks, pre-seed valuations should co-move with amounts raised while pre-seed ownership stakes should not. Second, and this is the paper’s central mechanism, the MFN clause makes the pre-seed investor the agent who decides who bears it. Pricing low hands the tranche a cheap conversion and keeps it inside the pre-seed block; pricing high pushes it into the priced round, where the seed investor must clear it. The two branches are governed by the *same* threshold equation evaluated at two effective investment costs, I and $I + m$, and the associated thresholds are strictly ordered, $\pi_1 < \pi_2$. That ordering is what makes the equilibrium partition well defined: on small needs the clause is threshold-neutral, on intermediate needs the pre-seed investor bunches exactly at the round price—a testable atom at $V_p/V = 1$ —and on large needs the tranche degenerates into a tax on the investment cost. Third, if the pre-seed investor holds a best-price SAFE rather than a fixed-base SAFE, its high-price branch disappears altogether, and the instrument is weakly dominated by a fixed-base SAFE, strictly so on large needs. The cap is a protection that guarantees its holder the better of two prices, and it delivers exactly that; but by loading the capitalization table in the states where the round prices low, it raises the financing threshold and destroys the states in which the claim converts at all. A protection granted to an investor can reduce that investor’s own payoff.

The rest of the paper proceeds as follows. Section 2 lays out the environment—agents, valuations, and the timing of the three financing decisions. Section 3 solves the pre-2022 contract by backward induction and establishes the benchmark against which the reform is measured, including the independence of the pre-seed share from the amount advanced and the partition of liquidity shocks into no-round, pre-seed and liquidation regions. Section 4 turns to the 2022 reform, separating its cash channel from its MFN channel, deriving the two conversion regimes and the ordering $\pi_1 < \pi_2$ on which the analysis rests, and comparing the pre-seed investor’s instruments. Section 5 draws the empirical implications and returns to the claims made for the reform. Section 6 concludes, and all proofs are collected in the Appendix.

2 Model

2.1 Agents

Suppose there is no time preference and all the agents below are risk-neutral and profit-maximizing.

Let's introduce entrepreneurs without initial budget. Each entrepreneur carries a project requiring a two-stage financing path: surviving an interim liquidity shock $l \in (0, 1)$ and funding a sunk investment cost I at a priced seed round. Each project yields a stochastic outcome in $\{0, 1\}$. 0 means failure and 1 means success. Without l , the project cannot reach the seed round. Without I at the seed round, the project cannot succeed. A project is characterized by its probability of success $\pi \in [0, 1]$. Note that the probability of success is equal to the expected outcome. Each entrepreneur knows the expected outcome of his project. Let f be the probability density function of π , F denote its cumulative distribution function, g be the probability density function of l , G denote its cumulative distribution function.

Monopolistic accelerators and pre-seed investors are sponsors for entrepreneurs to help them survive the interim liquidity shock until the revelation of their projects' expected outcome and the seed stage. Accelerators invest ex-ante liquidity shock. Pre-seed investors invest ex-post liquidity shock.

Investment cost I must be financed by monopolistic seed investors after the revelation of the project's expected outcome π .

2.2 Allocations and valuations

When seed investors allocate I to a project, it is with the promise that when payoffs are realized, they will take a *share* ρ_s of the realized outcome of the project. The *post-money valuation* V of a project corresponds to the ratio of the amount invested I by the share ρ_s at the seed stage. The *pre-money valuation* is $V - \sigma$.

Without knowing either π or l but their respective distributions, the accelerator invests σ_a in exchange of a share ρ_a . After the liquidity shock l , the pre-seed investor invests σ_p in exchange of a share ρ_p . The share ρ_p can be fixed at the time of pre-seed investment or can be conditioned on the seed stage.

2.3 Definitions and assumptions

A project's *initial net present value* is defined as the difference of its expected outcome and the sum of the expected liquidity shock and the investment cost. For notational

simplicity, the *initial market net present value*, which corresponds to the initial outside option's net present value, is taken to be 0.

(A1): $\mathbb{E}(\pi - l) - I > 0$ and f is log-concave

The first part of (A1) means that the average project is worth financing. This assumption can be thought of as conditioned on an accelerator preselection process supposed here independent from the game. The second part—that $\log f$ is concave—is a standard regularity condition satisfied by the uniform, normal, logistic, and exponential distributions, among others. It implies that the hazard rate $f/(1 - F)$ is increasing.

2.4 Timing

The game structure is as follows. First, without knowing either π or l but their respective distributions, the accelerator proposes (ρ_a, σ_a) to the entrepreneur. Then a liquidity shock occurs and the pre-seed investor proposes (ρ_p, σ_p) to cover the shock. Finally, a market shock reveals π and enables a seed round proposal of $(I, \rho_s) = (I, \frac{I}{V(\pi)})$.

The entrepreneur's outside option is normalized to zero at each stage.

3 Benchmark: fixed-equity SAFE

Before proceeding, it is useful to establish a benchmark result, namely that when the accelerator's contract is independent from the pre-seed and seed rounds, raising the amount invested increases seed success and the probability that no bridge is needed. I solve by backward induction.

3.1 Seed stage

The seed investor solves:

$$\max_V \int (\frac{I}{V(\pi)} \pi - I) f(\pi) d\pi$$

under: $\rho_a + \rho_p + \frac{I}{V(\pi)} \leq 1$

Proposition 1. *The seed investor proposes the valuation $V = \frac{I}{1 - \rho_a - \rho_p}$ to all the projects above $\pi_{\min} = \frac{I}{1 - \rho_a - \rho_p}$.*

Some worth financing projects don't reach the seed stage.

3.2 Pre-seed stage

The pre-seed investor moves at $t = 4$, after the liquidity shock l has been realised but before π is revealed. It must cover the residual need $\sigma_p = l - \sigma_a$ and chooses the valuation V_p at which its claim converts—or declines to participate. Its share is $\rho_p = \sigma_p/V_p$, so choosing V_p is equivalent to choosing ρ_p .

Throughout, write

$$M(x) := \int_x^1 \pi f(\pi) d\pi, \quad \text{so that} \quad M'(x) = -xf(x).$$

$M(x)$ is the expected outcome contributed by projects above x ; it is the object that governs every participation decision below.

Before solving for the price, one degree of freedom must be closed. Nothing so far prevents the pre-seed investor from injecting *more* than the residual need: it could advance any $x \geq l - \sigma_a$ and take a correspondingly larger share. It does not.

Lemma 1. *The pre-seed investor advances exactly the residual need, $\sigma_p = l - \sigma_a$.*

The economics is worth one sentence, because it is specific to this environment: money raised at the pre-seed stage does not reduce the sum the seed investor must put in. It only buys survival until the priced round. Over-funding would dilute the entrepreneur without lowering I , so it cannot improve the odds that a round happens.

With $\sigma_p = l - \sigma_a$ fixed, the pre-seed investor is repaid only on projects that reach and clear the seed round, that is on $\{\pi \geq \pi_{\min}\}$, and loses its outlay otherwise:

$$\max_{V_p} \int_{\pi_{\min}}^1 \left(\frac{\sigma_p}{V_p} \pi - \sigma_p \right) f(\pi) d\pi - \int_0^{\pi_{\min}} \sigma_p f(\pi) d\pi = \max_{V_p} \underbrace{\sigma_p \left(\frac{M(\pi_{\min}(V_p))}{V_p} - 1 \right)}_{=: \Pi_p(V_p)}.$$

Proposition 2. *The pre-seed investor proposes the valuation V_p^* satisfying*

$$\sigma_p f(\pi_{\min}) \pi_{\min}^3 = I V_p M(\pi_{\min}).$$

Corollary 1. *The optimal threshold $\pi_{\min}^*$ solves*

$$(1 - \rho_a) f(\pi) \pi^3 = I \left(M(\pi) + f(\pi) \pi^2 \right),$$

an equation in which σ_p does not appear. Consequently the equilibrium pre-seed share $\rho_p^ = 1 - \rho_a - I/\pi_{\min}^*$, the threshold $\pi_{\min}^*$, and the conditional seed probability $1 - F(\pi_{\min}^*)$ are all independent of the amount raised at the pre-seed stage, $\sigma_p = l - \sigma_a$.*

Two empirical implications follow. When a pre-seed round is needed at all, additional accelerator cash does not improve seed prospects *through the terms of that round*:

$\pi_{\min}^*$ is invariant to (σ_p, ρ_p) . And across companies facing different liquidity needs but sharing (ρ_a, I, F) , pre-seed *valuations* should co-move one-for-one with the amounts raised while pre-seed *ownership stakes* should not.

Proposition 3. *The pre-seed investor participates if and only if*

$$\sigma_p = l - \sigma_a \leq M(\pi_{\min}^*) \rho_p^* = M(\pi_{\min}^*) \left(1 - \rho_a - \frac{I}{\pi_{\min}^*}\right) =: \bar{\sigma}_p.$$

There is therefore a maximum liquidity shock, determined entirely by (ρ_a, I, F) , beyond which no pre-seed financing occurs at any price.

Proposition 4. *The realisation of l sorts every firm into one of three regimes:*

- (N) No pre-seed round, $l \leq \sigma_a$: here $(\rho_p, \sigma_p) = (0, 0)$, the seed investor finances if and only if $(1 - \rho_a)\pi \geq I$, that is $\pi \geq \pi_0 := I/(1 - \rho_a)$, and the conditional seed probability is $1 - F(\pi_0)$.
- (P) Pre-seed round, $\sigma_a < l \leq \sigma_a + \bar{\sigma}_p$: here $\rho_p = 1 - \rho_a - I/\pi_{\min}^*$, the seed investor finances if and only if $\pi \geq \pi_{\min}^*$, and the conditional seed probability is $1 - F(\pi_{\min}^*)$.
- (L) Liquidation, $l > \sigma_a + \bar{\sigma}_p$: the shock is not covered, the firm is liquidated at zero before the seed stage, and the conditional seed probability is 0.

Moreover $\pi_0 < \pi_{\min}^*$, since $\rho_p^* > 0$: surviving the shock through a pre-seed round is not free, and the dilution it costs raises the bar at the seed round.

Proposition 5. *The unconditional probability of seed success is*

$$\mathbb{P}(\text{seed}) = G(\sigma_a)(1 - F(\pi_0)) + (G(\sigma_a + \bar{\sigma}_p) - G(\sigma_a))(1 - F(\pi_{\min}^*)).$$

4 Adding a most-favored-nation tranche

Suppose the accelerator now invests σ_a against a headline share ρ_a^0 and, in addition, an amount m whose conversion price is

$$\min(V, V_p),$$

the more favourable of the seed price and the pre-seed price. This is a *most-favored-nation* clause: the tranche automatically adopts the best terms granted to any investor entering before the first priced round, and in this environment there is exactly one such investor—the one that covers the interim shock.

Such a contract has two distinct channels, which we treat in turn: a *cash* channel, since the accelerator's committed amount rises, and an *MFN* channel, since the term

$m/\min(V, V_p)$ enters the capitalization table. The two need not push in the same direction, and separating them is the purpose of what follows.

Y Combinator’s January 2022 contract is the leading instance of such a clause: the standard offer rose from \$125,000 for 7% to \$500,000, the additional \$375,000 being invested on an unbest-price SAFE carrying exactly this provision. We return to that case in Section 5.

4.1 The cash channel

Proposition 6. *Increasing σ_a raises the probability of seed success and raises the probability that no pre-seed round is needed.*

The cash channel is thus unambiguously favourable, and it operates entirely through the *extensive* margin—which firms need a pre-seed round at all—rather than through the terms of that round.

4.2 The MFN channel: two conversion regimes

At $t = 7$ the seed investor now solves

$$\max_V \int \left(\frac{I}{V(\pi)} \pi - I \right) f(\pi) d\pi \quad \text{subject to} \quad \rho_a^0 + \frac{m}{\min(V, V_p)} + \rho_p + \frac{I}{V} \leq 1.$$

As in the benchmark the objective is strictly increasing in the seed share, so the cap table binds. Which of the two prices the MFN tranche converts at determines the shape of the constraint, and hence the equilibrium valuation:

$$V = \begin{cases} \frac{I + m}{1 - \rho_a^0 - \frac{\sigma_p}{V_p}} & \text{if } V < V_p \quad (\text{the tranche converts at the round price}), \\ \frac{I}{1 - \rho_a^0 - \frac{\sigma_p + m}{V_p}} & \text{if } V > V_p \quad (\text{the tranche converts at the pre-seed price}). \end{cases}$$

Lemma 2. *Define*

$$\widehat{V}_p := \frac{I + m + \sigma_p}{1 - \rho_a^0}.$$

Then $V < V_p$ if and only if $V_p > \widehat{V}_p$, and $V > V_p$ if and only if $V_p < \widehat{V}_p$.

The economics is worth stating plainly. A pre-seed investor that prices the company *low* ($V_p < \widehat{V}_p$) hands the accelerator’s MFN tranche a cheap conversion: the tranche converts at V_p and lands in the pre-seed block of the cap table. A pre-seed investor that prices the company *high* ($V_p > \widehat{V}_p$) pushes the tranche into the seed round, where it converts at V and is borne by the seed investor. The pre-seed investor therefore chooses, through its own valuation, who pays for the accelerator’s clause.

4.3 The pre-seed investor's problem

The pre-seed investor's objective is unchanged in form, $\Pi_p(V_p) = \sigma_p(M(\pi_{\min}(V_p))/V_p - 1)$ with $\sigma_p = l - \sigma_a$, but $\pi_{\min}(\cdot)$ now depends on the regime. Since $\pi_{\min} = V$, Lemma 2 gives

$$\frac{\partial \pi_{\min}}{\partial V_p} = \begin{cases} -\frac{\pi_{\min}^2}{I+m} \cdot \frac{\sigma_p}{V_p^2} & \text{if } V_p > \widehat{V}_p, \\ -\frac{\pi_{\min}^2}{I} \cdot \frac{\sigma_p + m}{V_p^2} & \text{if } V_p < \widehat{V}_p, \end{cases}$$

and, proceeding exactly as in Proposition 2,

$$\Pi'_p(V_p) = \begin{cases} \frac{\sigma_p}{V_p^2} \left(\frac{\sigma_p f(\pi_{\min}) \pi_{\min}^3}{(I+m)V_p} - M(\pi_{\min}) \right) & \text{if } V_p > \widehat{V}_p, \\ \frac{\sigma_p}{V_p^2} \left(\frac{(\sigma_p + m) f(\pi_{\min}) \pi_{\min}^3}{I V_p} - M(\pi_{\min}) \right) & \text{if } V_p < \widehat{V}_p. \end{cases}$$

Substituting the cap-table identity of each regime— $(\sigma_p + m)/V_p = 1 - \rho_a^0 - I/\pi_{\min}$ in the low-price regime, $\sigma_p/V_p = 1 - \rho_a^0 - (I+m)/\pi_{\min}$ in the high-price regime—the two first-order conditions collapse onto a single family of equations, which is the observation the rest of the section rests on.

Definition 1. For $J > 0$ let

$$\Phi(\pi; J) := (1 - \rho_a^0) \pi^3 f(\pi) - J \left(\pi^2 f(\pi) + M(\pi) \right), \quad \pi \in (0, 1].$$

Lemma 3. The interior optimum of the pre-seed investor satisfies

$$\Phi(\pi_{\min}; I) = 0 \quad \text{in the low-price regime } V_p < \widehat{V}_p, \quad \Phi(\pi_{\min}; I+m) = 0 \quad \text{in the high-price regime } V_p > \widehat{V}_p$$

Lemma 3 is the pivot of the paper. The two regimes are governed by the *same* equation, evaluated at two different effective investment costs: I when the pre-seed investor absorbs the MFN tranche into its own block, and $I+m$ when it pushes the tranche into the seed round. Everything that follows is a comparison of the two roots.

Definition 2. Let π_1 denote the root of $\Phi(\cdot; I)$ and π_2 the root of $\Phi(\cdot; I+m)$, when they exist.

4.4 Existence, uniqueness, and the ordering of the thresholds

We first record a restriction without which no pre-seed round can occur at all.

Lemma 4. If $I+m \geq 1 - \rho_a^0$, the pre-seed investor never participates.

We therefore maintain $I + m < 1 - \rho_a^0$ throughout, and add one regularity condition.

(A2): $\pi \mapsto \pi^3 f(\pi)$ is non-decreasing on $(0, 1]$.

(A2) is weak: it holds for every non-decreasing density, and more generally whenever $3f(\pi) + \pi f'(\pi) \geq 0$. The uniform density satisfies it with $f' \equiv 0$.

Lemma 5. *Fix J with $0 < J < 1 - \rho_a^0$. Then $\Phi(\cdot; J)$ has a unique root π_J in $(J/(1 - \rho_a^0), 1)$, and $\Phi(\pi; J) < 0$ for $\pi < \pi_J$ and $\Phi(\pi; J) > 0$ for $\pi > \pi_J$ on that interval.*

We can now state and prove the comparison the model turns on.

Proposition 7. *Under (A1)–(A2) and $I + m < 1 - \rho_a^0$,*

$$\boxed{\pi_1 < \pi_2}$$

That is: the seed threshold is strictly higher when the pre-seed investor prices the company high—pushing the accelerator’s MFN tranche into the seed round—than when it prices it low and absorbs the tranche itself. When pre-seed investors overvalue the project, strictly fewer projects are financed at the seed round.

Remark 1. *The mechanism is transparent once the two regimes are read as the same problem at two different investment costs. In the low-price regime the pre-seed investor absorbs the MFN tranche into its own block: the seed investor still has to fund I , and the threshold is the one that solves $\Phi(\cdot; I) = 0$ —exactly the benchmark equation of Corollary 1, with ρ_a^0 in place of ρ_a . The MFN is then threshold-neutral. In the high-price regime the tranche is pushed into the seed round: the seed investor must now clear $I + m$ out of the same cap table, and the threshold rises to the root of $\Phi(\cdot; I + m)$.*

Example 1 (uniform density). *Let $\pi \sim U[0, 1]$, so $f \equiv 1$ and $M(x) = (1 - x^2)/2$. Then*

$$\Phi(\pi; J) = (1 - \rho_a^0)\pi^3 - \frac{J}{2}(\pi^2 + 1),$$

and (A2) holds since π^3 is increasing. With $\rho_a^0 = 0.07$ and $I = 0.35$ one obtains $\pi_1 \approx 0.643$; adding an MFN tranche of $m = 0.10$ gives $\pi_2 \approx 0.715$. The gap $\pi_2 - \pi_1$ is the share of projects that are financed when the pre-seed investor prices low and abandoned when it prices high. These are the values used in Figures 1 and 2, which are illustrative and are not a calibration.

4.5 Three regimes of liquidity need

Proposition 7 is what makes the equilibrium partition well defined: the two cutoffs it orders are exactly the boundaries of the pre-seed investor’s optimal pricing regions.

Proposition 8. Let $\sigma_p = l - \sigma_a > 0$. The pre-seed investor's optimal valuation is

$$V_p^* = \begin{cases} \frac{\sigma_p + m}{1 - \rho_a^0 - I/\pi_1} < V & (i) \text{ small needs: } \sigma_p \leq (1 - \rho_a^0)\pi_1 - (I + m), \\ \widehat{V}_p = \frac{I + m + \sigma_p}{1 - \rho_a^0} = V & (ii) \text{ intermediate needs: } (1 - \rho_a^0)\pi_1 - (I + m) < \sigma_p \leq (1 - \rho_a^0)\pi_2 - (I + m), \\ \frac{\sigma_p}{1 - \rho_a^0 - (I + m)/\pi_2} > V & (iii) \text{ large needs: } \sigma_p > (1 - \rho_a^0)\pi_2 - (I + m). \end{cases}$$

The seed threshold is π_1 in regime (i), $\widehat{V}_p$ in regime (ii), and π_2 in regime (iii). Regime (ii) is non-empty precisely because $\pi_1 < \pi_2$.

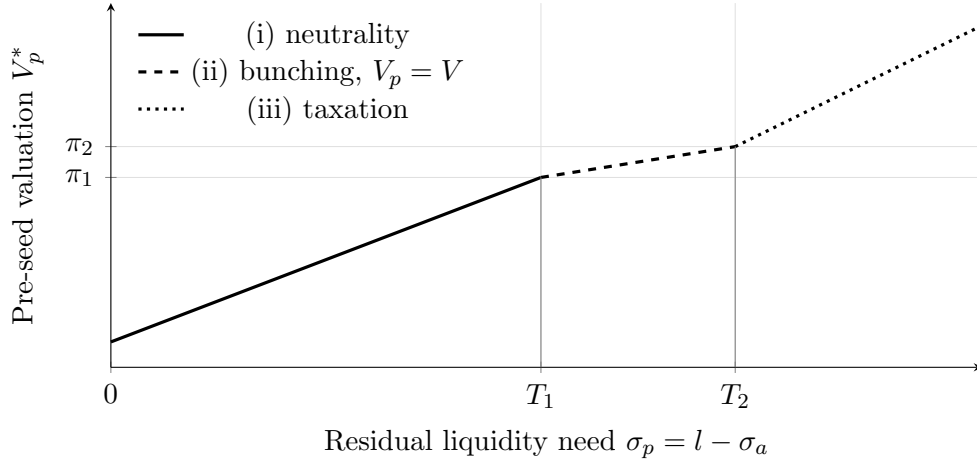


Figure 1: The pre-seed investor's optimal valuation across liquidity needs. On $[0, T_1]$ the investor reprices one-for-one and the financing threshold stays at π_1 ; on $[T_1, T_2]$ it bunches exactly at the round price; beyond T_2 it can no longer absorb the tranche and the threshold rises to π_2 . The middle interval has width $T_2 - T_1 = (1 - \rho_a^0)(\pi_2 - \pi_1)$ and is non-empty precisely because $\pi_1 < \pi_2$ (Proposition 7). Drawn for f uniform, $\rho_a^0 = 0.07$, $I = 0.35$, $m = 0.10$; the parameters are illustrative and are not a calibration.

Regime (ii) is the model's sharpest empirical prediction: over an interval of liquidity needs the pre-seed investor prices *exactly at the round price*, producing an atom at $V_p/V = 1$ in the cross-section of pre-seed valuations. The bunching is a direct consequence of the MFN clause, since $\widehat{V}_p$ is the price at which the tranche is indifferent between converting at the pre-seed price and at the round price.

4.6 When the pre-seed investor holds the same clause

So far the pre-seed investor has been assumed to hold a *fixed-base* SAFE: it negotiates a base V_p , buys $\rho_p = \sigma_p/V_p$, and holds that share whatever the round price turns out to

be. Suppose instead that it secures for itself the very protection the accelerator holds against it—a conversion at the more favourable of the two bases. Writing $b := \sigma_p$ for the amount advanced, its share becomes

$$\frac{b}{\min(V, V_p)},$$

which is exactly the form of the accelerator’s MFN tranche. In market terms this is the standard capped SAFE, whose holder converts at the lower of its cap and the round price; we call it a *best-price* SAFE to keep the mechanism, rather than the product name, in view.

The question this subsection answers is therefore not which of two products is better, but what happens when the clause is held by an investor who could otherwise have set its own price. The answer is that the protection is not free: on large financing needs it strictly hurts the investor holding it.

The seed investor’s problem. Both the tranche and the pre-seed claim now convert at $\min(V, V_p)$, so the saturated cap table reads

$$\rho_a^0 + \frac{m}{\min(V, V_p)} + \frac{b}{\min(V, V_p)} + \frac{I}{V} = 1,$$

which gives

$$V = \begin{cases} \frac{I + m + b}{1 - \rho_a^0} = \widehat{V}_p & \text{if } V < V_p \quad (\text{the cap does not bind}), \\ \frac{I}{1 - \rho_a^0 - \frac{m + b}{V_p}} & \text{if } V > V_p \quad (\text{the cap binds}), \end{cases}$$

and, exactly as in Lemma 2, the first branch obtains when $V_p \geq \widehat{V}_p$ and the second when $V_p < \widehat{V}_p$, with $\widehat{V}_p = (I + m + b)/(1 - \rho_a^0)$ unchanged.

The asymmetry with the fixed-base case is immediate and is the whole of the argument: above $\widehat{V}_p$ the round price V no longer depends on V_p . Once the cap stops binding, the pre-seed investor’s own pricing decision has no effect on anything.

The pre-seed investor’s problem. The objective is $\Pi_p(V_p) = b(M(\pi_{\min})/\min(V, V_p) - 1)$.

Below $\widehat{V}_p$ the cap binds, conversion occurs at V_p , and $\pi_{\min} = I/(1 - \rho_a^0 - (m + b)/V_p)$, so

$$\frac{\partial \pi_{\min}}{\partial V_p} = -\frac{\pi_{\min}^2}{I} \cdot \frac{b + m}{V_p^2} < 0, \quad \Pi'_p(V_p) = \frac{b}{V_p^2} \left(\frac{(b + m)f(\pi_{\min})\pi_{\min}^3}{I V_p} - M(\pi_{\min}) \right),$$

and substituting the cap-table identity $(m+b)/V_p = 1 - \rho_a^0 - I/\pi_{\min}$ into the first-order condition returns $\Phi(\pi_{\min}; I) = 0$ —the *same* equation as in the fixed-base low-price regime. The interior optimum is again π_1 .

Above $\widehat{V}_p$ the cap does not bind, conversion occurs at $V = \widehat{V}_p$, and both the price and the threshold are constants:

$$\Pi_p(V_p) = b \left(\frac{M(\widehat{V}_p)}{\widehat{V}_p} - 1 \right) \quad \text{for every } V_p \geq \widehat{V}_p,$$

so $\Pi'_p(V_p) = 0$ on that entire range. Any valuation above $\widehat{V}_p$ is optimal, and all of them deliver the same payoff.

Proposition 9. *Under a best-price SAFE the pre-seed investor's problem has only two regimes:*

(i) small needs, $b \leq (1 - \rho_a^0)\pi_1 - (I + m)$: *the cap binds,*

$$V_p^* = \frac{b + m}{1 - \rho_a^0 - I/\pi_1} < V, \quad \rho_p^* = \left(1 - \rho_a^0 - \frac{I}{\pi_1} \right) \frac{b}{b + m},$$

and the seed threshold is π_1 ;

(ii) large needs, $b > (1 - \rho_a^0)\pi_1 - (I + m)$: *the cap does not bind, every $V_p \geq \widehat{V}_p$ is optimal, and the seed threshold is $\widehat{V}_p = (I + m + b)/(1 - \rho_a^0)$.*

The contrast with Proposition 8 is the point. Under a fixed-base SAFE claim the pre-seed investor has three regimes and, on large needs, still optimises—it prices high, pushes the tranche into the round, and attains the threshold π_2 . Under a best-price SAFE the high-price branch collapses: the cap hands the round the conversion, the threshold jumps to $\widehat{V}_p$, and the investor loses the ability to price at all.

Proposition 10 (The clause hurts a holder who can price). *The best-price SAFE is weakly dominated by the fixed-base SAFE for the pre-seed investor, and strictly dominated on large needs, i.e. whenever $b > (1 - \rho_a^0)\pi_1 - (I + m)$. An investor who adopts the clause it is itself the target of is therefore weakly worse off, and strictly worse off exactly when the clause would bind.*

Remark 2 (Why a protection can hurt). *The economics runs against instinct. The clause is a guarantee: it ensures its holder converts on the better of the two bases, and it does. But conversion is not a transfer between the pre-seed investor and someone outside the company—it is a claim on the same capitalization table that must also accommodate I . By loading that table precisely in the states where the round prices low, the clause raises the threshold above which a round happens at all, and so destroys the states in which the claim pays anything. The protection operates through a cap-table externality that turns against its own holder. Waiving it is, in this sense, a*

commitment device: it leaves room for the seed investor and buys back the financing states.

Read against Section 1, this delivers the paper’s sharpest statement about the clause itself. The accelerator commits before anything is known and cannot revise its terms; for it, the clause is pure gain, since it recovers a base it could never have negotiated. The pre-seed investor arrives after the shock and sets a price; for it, the same clause substitutes a mechanical conversion rule for a decision it was making optimally, and the substitution can only weakly help. The value of a most-favored-nation clause is therefore not a property of the clause but of its holder: it protects an investor who cannot reprice, and constrains one who can.

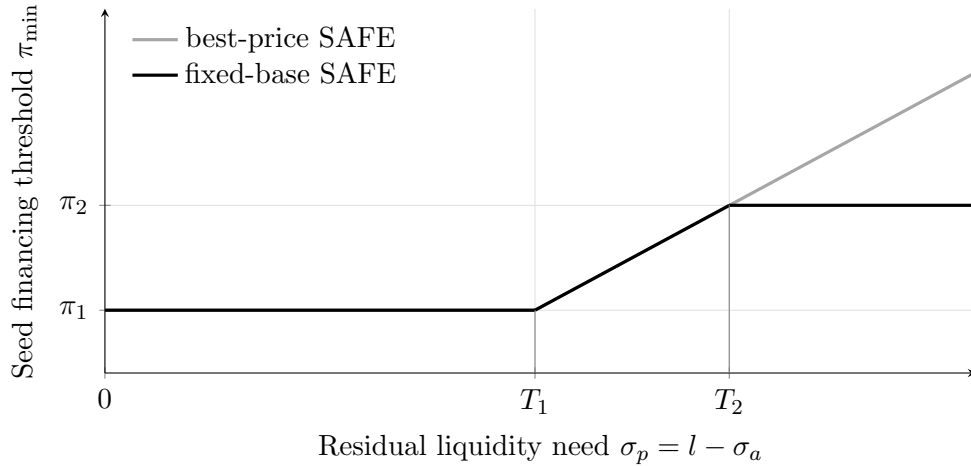


Figure 2: Financing thresholds under the two instruments. They coincide up to T_2 . Beyond it, the fixed-base investor prices high and holds the threshold at π_2 , whereas the best-price investor has no high-price branch: its claim converts at the round price, the threshold keeps rising with the shock, and the states in which the claim would have converted are destroyed. The vertical gap to the right of T_2 is the loss the clause inflicts on the investor holding it (Proposition 10). Same illustrative parameters as Figure 1.

Two empirical echoes follow. Distressed pre-seed rounds—those with large b , where Proposition 10 bites strictly—should migrate away from best-price SAFEs towards fixed-share instruments. And the dominance of SAFEs in ordinary times reads as the market revealing that typical needs sit in region (i), where the two instruments coincide exactly and the SAFE wins on transaction costs alone.

5 Discussion

5.1 Empirical implications

The model delivers four predictions that do not require knowing the primitives.

Valuations move with amounts, stakes do not. By Corollary 1, the pre-seed investor’s equilibrium share is pinned down by (ρ_a, I, F) alone. Across companies from the same cohort facing different liquidity shocks, the amount raised at the pre-seed stage and the valuation at which it is raised should move together one-for-one, while pre-seed ownership stakes should be flat in the amount. This is a within-cohort prediction and therefore does not require comparing accelerators to one another.

An atom at the round price. By Proposition 8, over an interval of liquidity needs the pre-seed investor prices exactly at the subsequent round’s valuation. In a cross-section of pre-seed rounds followed by a priced round, the distribution of V_p/V should therefore display a mass point at one. Bunching of this kind is a signature of the clause: $\hat{V}_p$ is precisely the price at which the tranche is indifferent between converting at the pre-seed price and at the round price, and nothing else in the environment selects it.

Instrument choice varies with distress. By Proposition 10, best-price SAFEs and fixed-base SAFEs are equivalent on small needs and strictly ranked on large ones. Pre-seed rounds raised under distress—large residual needs relative to the accelerator’s check—should therefore migrate away from capped instruments towards fixed-share ones, while the observed dominance of SAFEs in ordinary times is consistent with typical needs sitting in the region where the two coincide and the SAFE wins on transaction costs alone.

The net effect on seed success is ambiguous. The two channels of the reform pull in opposite directions. The cash channel raises the probability of reaching a priced round unambiguously (Proposition 6), by expanding the region in which no pre-seed round is needed at all. The MFN channel raises the financing threshold whenever the pre-seed investor prices high (Proposition 7) and, in the bunching region, fixes it at $\hat{V}_p$, which is increasing in the liquidity need. Which force dominates depends on F , G and m relative to typical needs. The sign is therefore an empirical question rather than a theoretical one, and the natural design compares seed-graduation rates of comparable cohorts around January 2022, using companies from accelerators that did not change their terms as a control group.

5.2 What the clause achieves, and where

It is worth separating what the clause is designed to do from what it does to financing, because the model shows that the two do not coincide.

The distributive objective is met in every regime. Whatever price the pre-seed investor sets, the tranche converts on the more favourable of the two bases, so the accelerator is never left holding terms worse than those obtained by the investor that came after it. That is what the clause was written for, and nothing in the analysis undoes it. In this sense the remedy works exactly as intended.

The effect on financing depends on whether the pre-seed investor can absorb the clause in its own price. On small needs it can. It raises its valuation by exactly the tranche, the total convertible share it targets is unchanged, and the financing threshold is the one that would have prevailed without the clause (Lemma 3 and Proposition 8). The accelerator gains equity, the pre-seed investor gives up precisely that equity, and no project that would otherwise have been financed fails to be. The clause is purely redistributive.

On large needs it cannot. The price the pre-seed investor would need to set is pinned down by the size of the shock it must cover, the tranche is pushed into the priced round, and the seed investor must clear $I+m$ out of the same capitalization table. The threshold rises from π_1 to π_2 , and the projects lying between them—worth financing, and financed in the absence of the clause—are abandoned. Here the accelerator's equity is obtained at the cost of companies that never reach a round at all.

A clause of this kind is therefore harmless precisely where it is redistributive, and costly precisely where it binds. The design implication is not that most-favored-nation clauses are undesirable, but that their cost falls on the firms with the largest interim funding needs—which are, by construction, the firms in the weakest position.

5.3 What the model does and does not say about the reform

Of the three arguments made for the reform, two are addressed by the analysis and one is not.

The claim that the reform reduces fundraising pressure is supported, through two distinct mechanisms that the model separates. The cash channel mechanically enlarges the set of firms that never need an interim round. Conditional on needing one, the clause raises the valuation at which it is raised by exactly the tranche in the neutrality region, while lowering the pre-seed investor's stake to $\rho_p^* \sigma_p / (\sigma_p + m)$. Less dilution at a higher price is what the argument asserted, and it holds—but only where the pre-seed investor can reprice, which is the small-need region.

The claim about seed success is ambiguous in the model, as set out above, and the

ambiguity is a result rather than an evasion: it identifies the conditions under which the reform helps and those under which it does not.

The claim that the reform attracts more founders lies outside the model. The environment closes with the seed investor extracting the residual, so the entrepreneur’s ex-ante value is zero both before and after the reform, and the margin along which additional applicants would be attracted does not exist here. Evaluating that claim requires a closure with positive founder rents—a competitive seed stage, for instance—under which the founder’s ex-ante value would inherit the same regime structure. We do not model it and make no claim about it.

5.4 Limitations

Three modelling choices bound the analysis. The pre-seed investor is a monopolist: under competition its profit is driven to zero and its valuation is pinned at the participation frontier $M(\pi_{\min}) = V_p$ rather than by the first-order condition, which reallocates rents without altering the regime structure. The seed investor likewise extracts the residual equity, which is what makes the founder margin unavailable. And the liquidity shock is exogenous: a firm anticipating the clause might adjust its burn rate, which would make l a choice variable and the regimes endogenous to the contract itself. Each of these extensions preserves the architecture—a threshold, a participation cap, and the two-branch structure of the clause—while changing who captures the surplus.

6 Conclusion

We have studied what a most-favored-nation clause in an accelerator’s contract does to the financing round that follows it. The clause is written to protect the accelerator against terms granted to later investors, and it does so mechanically. Its equilibrium effect, however, runs through an agent it does not bind: the pre-seed investor, who decides by its own pricing whether the tranche converts inside its block or inside the priced round.

That decision is governed by a single threshold equation evaluated at two effective investment costs, and the resulting thresholds are strictly ordered. The ordering partitions liquidity needs into three regimes—neutrality, bunching, taxation—which are not artefacts of a parametrisation but consequences of the clause interacting with a capitalization table that must also accommodate the seed investment. Where the pre-seed investor can reprice, the clause leaves the financing threshold of the chain untouched and raises valuations by exactly the tranche. Where it cannot, the tranche becomes a tax on the investment cost and fewer projects are financed.

The instrument comparison sharpens the same point. A best-price SAFE guarantees

its holder the better of the pre-seed price and the round price, and it delivers precisely that. It also removes that holder's ability to price, and loads the capitalization table exactly in the states where the round prices low, raising the threshold above which any round occurs. On large financing needs the protection is strictly worse for the investor it protects than a claim carrying no protection at all. Waiving the cap is a commitment device.

Taken together, the two halves of the paper say something about the clause that neither says alone. A most-favored-nation provision has no intrinsic value: its worth depends on the position of the investor holding it. The accelerator commits at admission, before anything is known, on terms it can no longer revise; for it the clause is a pure gain, since it recovers a base it would never have been able to negotiate. The pre-seed investor arrives after the shock and *sets* a price; for it the same clause replaces a decision it was taking optimally with a mechanical conversion rule, and can therefore only weakly help. The clause protects an investor who cannot reprice and constrains one who can.

This is why sequence, and not the contract alone, determines the effect. Contractual protection granted at one financing stage is never neutral for the stages that follow, because the investors at those stages adjust the terms they offer; and the incidence of the protection—who ultimately bears it—is endogenous to the size of the financing need rather than fixed by the clause. Whether a reform of this kind improves the financing prospects of the companies it is designed to help is, for that reason, a question the theory poses sharply and leaves to the data.

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A Proofs

Proof of Proposition 1

Proof. The objective is strictly increasing in the share $\frac{I}{V}$, so the cap-table constraint binds; and the seed investor proposes if and only if its payoff is non-negative. $\square$

Proof of Lemma 1

Proof. Parameterise the pre-seed investor's offer by the pair (x, ρ_p) , where x is the amount advanced and ρ_p the share obtained; this is equivalent to choosing (x, V_p) with $\rho_p = x/V_p$, and since V_p is unrestricted the two components can be set independently. The investor is repaid $\rho_p \pi$ on projects that clear the seed round and loses x otherwise, so its expected payoff is

$$\Pi_p(x, \rho_p) = \rho_p M(\pi_{\min}(\rho_p)) - x.$$

The seed threshold depends on the *capitalization table* and not on the cash advanced: $\pi_{\min} = I/(1 - \rho_a - \rho_p)$ by Proposition 1, a function of ρ_p alone. Hence at fixed ρ_p , $\partial \Pi_p / \partial x = -1 < 0$: additional cash is pure cost, since it buys no additional equity at a given share and does not relax the constraint that binds at the seed round. The firm is liquidated at zero if the shock is not covered, so any $x < l - \sigma_a$ yields $-x$ with certainty and is dominated by not participating. The optimum is therefore $x^* = l - \sigma_a$. $\square$

Proof of Proposition 2

Proof. By Proposition 1, $\pi_{\min} = I/(1 - \rho_a - \rho_p)$ with $\rho_p = \sigma_p/V_p$, so

$$\frac{\partial \pi_{\min}}{\partial V_p} = -\frac{\pi_{\min}^2}{I} \cdot \frac{\sigma_p}{V_p^2} < 0 :$$

a more generous pre-seed valuation dilutes the entrepreneur less, leaves more room in the cap table for the seed investor, and therefore lowers the seed threshold. Differentiating Π_p and using $M'(x) = -xf(x)$,

$$\Pi'_p(V_p) = \sigma_p \left(\frac{M'(\pi_{\min}) \partial \pi_{\min} / \partial V_p}{V_p} - \frac{M(\pi_{\min})}{V_p^2} \right) = \frac{\sigma_p}{V_p^2} \left(\frac{\sigma_p f(\pi_{\min}) \pi_{\min}^3}{I V_p} - M(\pi_{\min}) \right).$$

Setting $\Pi'_p(V_p) = 0$ yields the stated first-order condition. $\square$

Proof of Corollary 1

Proof. Substitute $V_p = \sigma_p / \rho_p$ into Proposition 2: the factor σ_p cancels on both sides, leaving $f(\pi) \pi^3 \rho_p = I M(\pi)$. Replacing $\rho_p = 1 - \rho_a - I/\pi$ and rearranging gives the stated equation. $\square$

Proof of Proposition 3

Proof. Participation requires $\Pi_p(V_p^*) \geq 0$, i.e. $M(\pi_{\min}^*) \geq V_p^*$. Multiplying both sides by $\rho_p^* > 0$ and using $V_p^* \rho_p^* = \sigma_p$ gives the condition; ρ_p^* and $\pi_{\min}^*$ are the constants of Corollary 1. $\square$

Proof of Proposition 5

Proof. Condition on the three regimes of Proposition 4 and use $\mathbb{P}(\text{seed}) = \sum_R \mathbb{P}(R) \mathbb{P}(\text{seed} | R)$, with $\mathbb{P}(\text{seed} | L) = 0$. $\square$

Proof of Proposition 6

Proof. By Corollary 1, raising σ_a leaves π_0 , $\pi_{\min}^*$, ρ_p^* and $\bar{\sigma}_p$ unchanged: they depend on (ρ_a, I, F) only. What changes is the partition of the shock space in Proposition 4: the no-pre-seed region $\{l \leq \sigma_a\}$ expands and the liquidation region $\{l > \sigma_a + \bar{\sigma}_p\}$ shrinks, both by the same displacement. Differentiating Proposition 5,

$$\frac{\partial \mathbb{P}(\text{seed})}{\partial \sigma_a} = g(\sigma_a)(F(\pi_{\min}^*) - F(\pi_0)) + g(\sigma_a + \bar{\sigma}_p)(1 - F(\pi_{\min}^*)) > 0,$$

where the first term is non-negative because $\pi_0 < \pi_{\min}^*$ (Proposition 4) and the second is strictly positive. $\square$

Proof of Lemma 2

Proof. Take the first branch. $V < V_p$ reads $(I + m)/(1 - \rho_a^0 - \sigma_p/V_p) < V_p$, and since the denominator is positive whenever a seed round occurs, this is equivalent to $I + m < (1 - \rho_a^0)V_p - \sigma_p$, that is to $V_p > (I + m + \sigma_p)/(1 - \rho_a^0)$. The second branch is the same computation with m moved into the pre-seed block: $V > V_p$ reads $I > (1 - \rho_a^0)V_p - (\sigma_p + m)$, i.e. $V_p < \hat{V}_p$. $\square$

Proof of Lemma 3

Proof. In the low-price regime, set $\Pi'_p = 0$: $(\sigma_p + m)f(\pi)\pi^3 = IV_p M(\pi)$. Divide by V_p and use $(\sigma_p + m)/V_p = 1 - \rho_a^0 - I/\pi$:

$$\left(1 - \rho_a^0 - \frac{I}{\pi}\right)f(\pi)\pi^3 = I M(\pi) \iff (1 - \rho_a^0)\pi^3 f(\pi) - I\pi^2 f(\pi) - IM(\pi) = 0,$$

which is $\Phi(\pi; I) = 0$. In the high-price regime the same steps with $\sigma_p/V_p = 1 - \rho_a^0 - (I + m)/\pi$ and the factor $(I + m)$ in place of I give $\Phi(\pi; I + m) = 0$. $\square$

Proof of Lemma 4

Proof. Suppose a pre-seed round occurs. If $V_p > \widehat{V}_p$ then $\pi_{\min} = (I + m)/(1 - \rho_a^0 - \sigma_p/V_p) \geq (I + m)/(1 - \rho_a^0) \geq 1$, which is impossible since $\pi_{\min} \leq 1$. If $V_p \leq \widehat{V}_p$ then $\pi_{\min} = I/(1 - \rho_a^0 - (\sigma_p + m)/V_p)$ is decreasing in V_p , hence

$$\pi_{\min} \geq \pi_{\min}(\widehat{V}_p) = \frac{I}{1 - \rho_a^0 - \frac{(\sigma_p + m)(1 - \rho_a^0)}{I + m + \sigma_p}} = \frac{I + m + \sigma_p}{1 - \rho_a^0} \geq \frac{I + m}{1 - \rho_a^0} \geq 1,$$

again impossible. Hence $I + m < 1 - \rho_a^0$ whenever a pre-seed round occurs. $\square$

Proof of Lemma 5

Proof. Sign at the endpoints. At $\pi = J/(1 - \rho_a^0)$ the first two terms of Φ cancel, since $(1 - \rho_a^0)\pi^3 f - J\pi^2 f = \pi^2 f((1 - \rho_a^0)\pi - J) = 0$, leaving $\Phi = -J M(\pi) < 0$ because $M > 0$ on $(0, 1)$. At $\pi = 1$ we have $M(1) = 0$, so $\Phi(1; J) = f(1)((1 - \rho_a^0) - J) > 0$ by assumption on J . By continuity a root exists in between.

Uniqueness. Let π^* be any root, so that $J = (1 - \rho_a^0)\pi^{*3}f(\pi^*)/(\pi^{*2}f(\pi^*) + M(\pi^*))$. Differentiating Φ and using $M' = -\pi f$,

$$\frac{\partial \Phi}{\partial \pi} = (1 - \rho_a^0)\pi^2(3f + \pi f') - J\pi(f + \pi f').$$

Substituting the expression for J at π^* and clearing the positive denominator, the sign of $\partial \Phi / \partial \pi$ at π^* is the sign of

$$(3f + \pi^* f')(\pi^{*2}f + M) - \pi^{*2}f(f + \pi^* f') = 2\pi^{*2}f^2 + M(3f + \pi^* f'),$$

all functions evaluated at π^* . Under (A2) we have $3f + \pi f' \geq 0$, and $f > 0$, $M > 0$ on $(0, 1)$, so this expression is strictly positive. Every root is therefore a strict upcrossing, which forces the root to be unique and yields the stated sign pattern. $\square$

Proof of Proposition 7

Proof. The proof has two steps: $\Phi(\cdot; I)$ dominates $\Phi(\cdot; I+m)$ pointwise, and $\Phi(\cdot; I)$ crosses zero from below.

Step 1 (strict dominance). For every $\pi \in (0, 1)$,

$$\Phi(\pi; I) - \Phi(\pi; I+m) = -I(\pi^2 f + M) + (I+m)(\pi^2 f + M) = m(\pi^2 f(\pi) + M(\pi)) > 0,$$

since $m > 0$, $f > 0$ and $M > 0$ on $(0, 1)$. Note that $J \mapsto \Phi(\pi; J)$ is affine and strictly decreasing for each fixed π , which is the substance of this step: raising the effective investment cost lowers Φ everywhere.

Step 2 (evaluation at π_2). By definition $\Phi(\pi_2; I+m) = 0$. Applying Step 1 at $\pi = \pi_2$,

$$\Phi(\pi_2; I) = \Phi(\pi_2; I+m) + m(\pi_2^2 f(\pi_2) + M(\pi_2)) = m(\pi_2^2 f(\pi_2) + M(\pi_2)) > 0.$$

Step 3 (conclusion). By Lemma 5 applied with $J = I$, the function $\Phi(\cdot; I)$ is strictly negative to the left of its unique root π_1 and strictly positive to the right. Since $\Phi(\pi_2; I) > 0$ by Step 2, the point π_2 must lie strictly to the right of π_1 , i.e. $\pi_1 < \pi_2$.

Both roots exist and lie in $(0, 1)$ because $I < I+m < 1 - \rho_a^0$, so Lemma 5 applies to $J = I$ and to $J = I+m$. $\square$

Proof of Proposition 8

Proof. By Lemma 2 the low-price regime requires $V_p < \widehat{V}_p$, which given $\pi_{\min} = \pi_1$ is equivalent to $\pi_1 \geq \widehat{V}_p$, i.e. $(1 - \rho_a^0)\pi_1 \geq I+m+\sigma_p$: this is case (i). Symmetrically the high-price regime requires $\pi_2 < \widehat{V}_p$, i.e. $\sigma_p > (1 - \rho_a^0)\pi_2 - (I+m)$: this is case (iii). For σ_p strictly between the two bounds neither interior solution is admissible in its own regime, so the optimum is at the kink: Π_p is increasing to the left of $\widehat{V}_p$ and decreasing to its right, and $V_p^* = \widehat{V}_p$. The intervals of cases (i) and (iii) are disjoint and exhaust the line only if $(1 - \rho_a^0)\pi_1 - (I+m) < (1 - \rho_a^0)\pi_2 - (I+m)$, that is if and only if $\pi_1 < \pi_2$, which is Proposition 7. $\square$

Proof of Proposition 9

Proof. On $[0, \widehat{V}_p)$ the first-order condition is $\Phi(\pi_{\min}; I) = 0$, whose unique root is π_1 by Lemma 5; this interior solution is admissible precisely when the induced V_p^* satisfies $V_p^* < \widehat{V}_p$, equivalently $\pi_1 \geq \widehat{V}_p$, equivalently $b \leq (1 - \rho_a^0)\pi_1 - (I+m)$, which is case (i). Otherwise the constrained optimum lies on $[\widehat{V}_p, \infty)$, where Π_p is constant by the computation above, giving case (ii). $\square$

Proof of Proposition 10

Proof. Step 1: the fixed-base SAFE can replicate the best-price SAFE. Suppose the pre-seed investor holds a fixed-base SAFE and chooses $V_p = \widehat{V}_p$. Its share is then $b/\widehat{V}_p = b(1 - \rho_a^0)/(I + m + b)$, so the residual left to the seed investor is

$$1 - \rho_a^0 - \frac{b}{\widehat{V}_p} = (1 - \rho_a^0) \left(1 - \frac{b}{I + m + b} \right) = \frac{(1 - \rho_a^0)(I + m)}{I + m + b},$$

and the round price solves $V = (I + m)/(1 - \rho_a^0 - b/\widehat{V}_p) = (I + m + b)/(1 - \rho_a^0) = \widehat{V}_p$. Both the conversion price and the threshold therefore coincide with those of the best-price SAFE in its non-binding regime, and so does the payoff, $b(M(\widehat{V}_p)/\widehat{V}_p - 1)$. The capped-SAFE outcome is thus contained in the fixed-base investor's choice set, which gives weak dominance.

Step 2: strictness on large needs. On $b > (1 - \rho_a^0)\pi_1 - (I + m)$ the capped SAFE delivers the payoff of Step 1. The fixed-base investor, facing the same b , is not restricted to $V_p = \widehat{V}_p$: by Proposition 8 its optimum is either the bunching solution or the interior solution with threshold π_2 , and in the latter case $\pi_2 < \widehat{V}_p$ (this is exactly the defining inequality of regime (iii), $b > (1 - \rho_a^0)\pi_2 - (I + m)$). Since Π_p is strictly decreasing in the threshold—raising $\pi_{\min}$ destroys, one for one, the states in which the claim converts—and the fixed-base optimum attains a strictly lower threshold than $\widehat{V}_p$ at a price it chooses, its payoff is strictly higher. Hence strict dominance. $\square$